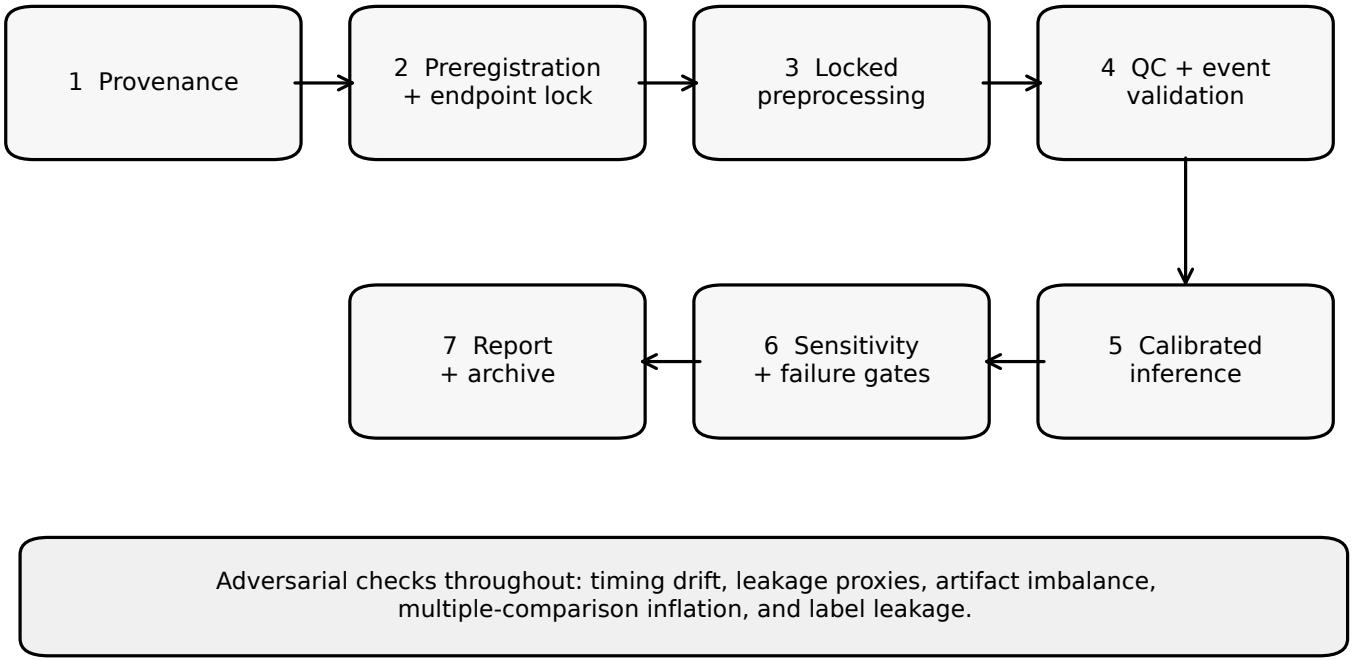**Retrospective reanalysis should move from provenance to inference through a locked, auditable pipeline.**